

```
In [1]: import pandas as pd
```

```
In [2]: df =  
pd.read_csv('C:\\Users\\dipanshus\\Downloads\\customer_shopping_behavior.csv')  
df.head()
```

Out[2]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location
0	1	55	Male	Blouse	Clothing	53	Kentucky
1	2	19	Male	Sweater	Clothing	64	Maine
2	3	50	Male	Jeans	Clothing	73	Massachusetts
3	4	21	Male	Sandals	Footwear	90	Rhode Island
4	5	45	Male	Blouse	Clothing	49	Oregon

```
In [7]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   str
3   Item Purchased                      3900 non-null   str
4   Category                            3900 non-null   str
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   str
7   Size                                3900 non-null   str
8   Color                               3900 non-null   str
9   Season                              3900 non-null   str
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   str
12  Shipping Type                      3900 non-null   str
13  Discount Applied                   3900 non-null   str
14  Promo Code Used                    3900 non-null   str
15  Previous Purchases                 3900 non-null   int64
16  Payment Method                     3900 non-null   str
17  Frequency of Purchases             3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

```
In [8]: df.describe(include='all')
```

Out[8]:

	Customer ID	Age	Gender	Item Purchased	Category	
count	3900.000000	3900.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN
top	NaN	NaN	Male	Blouse	Clothing	NaN
freq	NaN	NaN	2652	171	1737	NaN
mean	1950.500000	44.068462	NaN	NaN	NaN	59.0
std	1125.977353	15.207589	NaN	NaN	NaN	23.0
min	1.000000	18.000000	NaN	NaN	NaN	20.0
25%	975.750000	31.000000	NaN	NaN	NaN	39.0
50%	1950.500000	44.000000	NaN	NaN	NaN	60.0
75%	2925.250000	57.000000	NaN	NaN	NaN	81.0
max	3900.000000	70.000000	NaN	NaN	NaN	100.0



```
In [11]: df.isnull().sum()
```

```
Out[11]: Customer ID      0
         Age              0
         Gender           0
         Item Purchased    0
         Category         0
         Purchase Amount (USD) 0
         Location         0
         Size             0
         Color            0
         Season           0
         Review Rating    37
         Subscription Status 0
         Shipping Type     0
         Discount Applied  0
         Promo Code Used   0
         Previous Purchases 0
         Payment Method    0
         Frequency of Purchases 0
         dtype: int64
```

```
In [9]: # Grouping Review Rating on the basis of Category and then filling the
missing value with the median.
```

```
df['Review Rating'] = df.groupby('Category')['Review
Rating'].transform(lambda x: x.fillna(x.median()))
```

```
In [10]: df.isnull().sum()
```

```
Out[10]: Customer ID      0
Age      0
Gender    0
Item Purchased  0
Category  0
Purchase Amount (USD)  0
Location  0
Size      0
Color     0
Season    0
Review Rating  0
Subscription Status  0
Shipping Type  0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method  0
Frequency of Purchases  0
dtype: int64
```

```
In [11]: df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
```

```
In [12]: df.columns
```

```
Out[12]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
'purchase_amount_(usd)', 'location', 'size', 'color',
'season',
'review_rating', 'subscription_status', 'shipping_type',
'discount_applied', 'promo_code_used', 'previous_purchases',
'payment_method', 'frequency_of_purchases'],
dtype='str')
```

```
In [13]: df = df.rename(columns = {'purchase_amount_(usd)': 'purchase_amount'})
```

```
In [14]: df.columns
```

```
Out[14]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
              'purchase_amount', 'location', 'size', 'color', 'season',  
              'review_rating', 'subscription_status', 'shipping_type',  
              'discount_applied', 'promo_code_used', 'previous_purchases',  
              'payment_method', 'frequency_of_purchases'],  
              dtype='str')
```

```
In [15]: # create a column age_group  
labels = ['Yound Adult','Adult','Middle-aged','Senior']  
df['age_group'] = pd.qcut(df['age'],q = 4, labels = labels)  
df[['age','age_group']].head(10)
```

```
Out[15]:
```

	age	age_group
0	55	Middle-aged
1	19	Yound Adult
2	50	Middle-aged
3	21	Yound Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Yound Adult
8	26	Yound Adult
9	57	Middle-aged

```
In [17]: # create column purchase_frequency_days  
  
frequency_mapping = {  
    'Fortnightly': 14,  
    'Weekly': 7,  
    'Monthly': 30,  
    'Quarterly': 90,  
    'Bi-Weekly': 14,  
    'Annually' : 365,  
    'Every 3 Months' : 90  
}  
  
df['purchase_frequency_days'] =  
df['frequency_of_purchases'].map(frequency_mapping)
```

```
In [18]: df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

Out[18]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

```
In [24]: df[['discount_applied', 'promo_code_used']].head(10)
```

Out[24]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
In [19]: (df['discount_applied'] == df['promo_code_used']).all()
```

Out[19]: np.True_

```
In [20]: df = df.drop('promo_code_used', axis = 1)
df.columns
```

```
Out[20]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
               'purchase_amount', 'location', 'size', 'color', 'season',
               'review_rating', 'subscription_status', 'shipping_type',
               'discount_applied', 'previous_purchases', 'payment_method',
               'frequency_of_purchases', 'age_group',
               'purchase_frequency_days'],
              dtype='str')
```

```
In [27]: pip install psycpg2-binary sqlalchemy
```

```
Collecting psycogp2-binary
  Downloading psycogp2_binary-2.9.11-cp314-cp314-
win_amd64.whl.metadata (5.1 kB)
Collecting sqlalchemy
  Downloading sqlalchemy-2.0.49-cp314-cp314-win_amd64.whl.metadata
(9.8 kB)
Collecting greenlet>=1 (from sqlalchemy)
  Downloading greenlet-3.4.0-cp314-cp314-win_amd64.whl.metadata (3.8
kB)
Requirement already satisfied: typing-extensions>=4.6.0 in
C:\Users\dipanshus\AppData\Local\Programs\Python\Python314\Lib\site-
packages (from sqlalchemy) (4.15.0)
Downloading psycogp2_binary-2.9.11-cp314-cp314-win_amd64.whl (2.8 MB)
----- 0.0/2.8 MB ? eta -:-:--
----- 1.3/2.8 MB 8.5 MB/s eta
0:00:01
----- 2.8/2.8 MB 7.9 MB/s
0:00:00
Downloading sqlalchemy-2.0.49-cp314-cp314-win_amd64.whl (2.1 MB)
----- 0.0/2.1 MB ? eta -:-:--
----- 2.1/2.1 MB 10.8 MB/s eta
0:00:01
----- 2.1/2.1 MB 9.4 MB/s
0:00:00
Downloading greenlet-3.4.0-cp314-cp314-win_amd64.whl (239 kB)
Installing collected packages: psycogp2-binary, greenlet, sqlalchemy
```


[illegible]

```
----- 2/3 [sqlalchemy]
----- 3/3 [sqlalchemy]
```

Successfully installed greenlet-3.4.0 psycpg2-binary-2.9.11
sqlalchemy-2.0.49
Note: you may need to restart the kernel to use updated packages.

```
In [1]: from sqlalchemy import create_engine
```

```
In [3]: !pip install sqlalchemy pymysql
```

```
Requirement already satisfied: sqlalchemy in
C:\Users\dipanshus\AppData\Local\Programs\Python\Python314\Lib\site-
packages (2.0.49)
Collecting pymysql
  Downloading pymysql-1.1.2-py3-none-any.whl.metadata (4.3 kB)
Requirement already satisfied: greenlet>=1 in
C:\Users\dipanshus\AppData\Local\Programs\Python\Python314\Lib\site-
packages (from sqlalchemy) (3.4.0)
Requirement already satisfied: typing-extensions>=4.6.0 in
C:\Users\dipanshus\AppData\Local\Programs\Python\Python314\Lib\site-
packages (from sqlalchemy) (4.15.0)
Downloading pymysql-1.1.2-py3-none-any.whl (45 kB)
Installing collected packages: pymysql
Successfully installed pymysql-1.1.2
```

```
In [22]: # Step 1 : Connect to MySQL
# Replace placeholders with your actual details
username = "root"
password = "root"
hostname = "localhost"
port = "3306"
database = "customer_behavior"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{hostname}:
{port}/{database}")

# Step 2 : Load Dataframe into MySQL
table_name = "customer"
df.to_sql(table_name,engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database
'{database}'.")
```

Data successfully loaded into table 'customer' in database
'customer_behavior'.

```
In [24]: pd.read_sql("Select * From customer LIMIT 5;",engine)
```

Out[24]:

	customer_id	age	gender	item_purchased	category	purchase_arr
0	1	55	Male	Blouse	Clothing	53
1	2	19	Male	Sweater	Clothing	64
2	3	50	Male	Jeans	Clothing	73
3	4	21	Male	Sandals	Footwear	90
4	5	45	Male	Blouse	Clothing	49



```
In [ ]:
```